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Laminar Boundary Layer Over a Flat Plate

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1. Introduction

Let us consider air flowing under zero pressure gradient over a horizontal plate, forming a boundary layer above it. The inlet velocity of the air, at the start of the plate, is assumed uniform and parallel to its surface, with a value of 2 m/s. The equations describing this flow for zero pressure gradient are the Prandtl boundary layer equations.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{1.1}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = v\frac{\partial u^2}{\partial y^2} \tag{1.2}$$

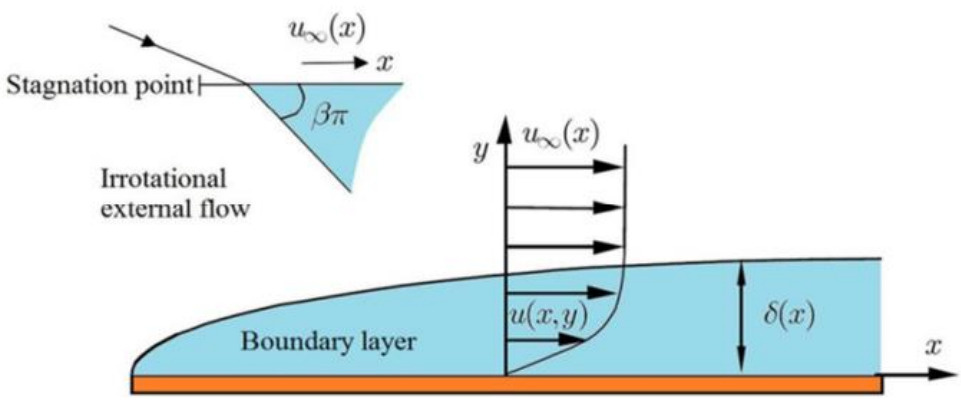

Figure 1.1. *Two-dimensional representation of boundary layer*

The objective of this study is to compute the air boundary layer over the plate by selecting an appropriate mesh, for a plate length of 10 meters.

2. Analytical Solution Calculation

The laminar boundary layer equations form a system of partial differential equations that can be solved numerically, but under specific conditions, an analytical solution can also be derived. In this case, for incompressible flow over a flat plate and with zero pressure gradient, the Blasius solution can be obtained.

The quantities derived from the Blasius equations are:

- Boundary layer thickness δ
- Displacement thickness δ_1
- Momentum thickness δ_2 for laminar boundary layer

as given by the following equations.

$$\delta = \frac{5x}{\sqrt{Re_x}} \ , \delta_1 = \frac{1.72x}{\sqrt{Re_x}} \ , \delta_2 = \frac{0.664x}{\sqrt{Re_x}} \ , C_f = \frac{0.664}{\sqrt{Re_x}}$$

Considering the air properties at ambient temperature, the Reynolds number is calculated as:

$$Re_x = \frac{u_\infty x}{v}$$

όπου $\begin{cases} u_\infty = 2 \ \frac{m}{s} \\ v = 1.48 * 10^{-5} \frac{m^2}{s} \end{cases}$

Thus, the first comparison values of the required quantities along the plate are obtained. Indicatively, at the mid-point of the plate, the following values arise.

Variables	Values
δ [mm]	0.03
δ_1 [mm]	0.0105
δ_2 [mm]	0.00404
C_f	0.0008

Table 2.1. *Results from the analytical solution at the mid-point of the plate*

3. Selection of Discretization Methods

Up to this point, we have analytically calculated the quantities characterizing the boundary layer by solving the boundary layer equations. However, the range of problems that can be addressed by these methods is limited, as this approach relies on several assumptions. To expand the ability to solve boundary layer equations, we consider numerical solutions.

3.1 Explicit Method

In Computational Fluid Dynamics (CFD), an explicit method refers to a numerical scheme for solving the governing equations of fluid flow, where the solution is computed directly in time. In explicit methods, the solution at the next timestep is calculated directly from known information at the current step, without solving a system of equations.

In terms of time integration, explicit methods are contrasted with implicit methods. In an explicit method, the solution for the unknown variables is obtained straightforwardly, often through a finite difference approach, without solving algebraic systems at each time step. However, explicit methods may have stability limitations that restrict the time step size.

The basic idea is to separate the spatial and temporal domains and then update the solution at each grid point step-by-step using information from neighboring points. The direct nature of the method allows simplicity in implementation but may come at the cost of stability restrictions.

In this example, first-order derivatives are approximated using forward, backward, and central differences.

3.2 Implicit Method

In CFD, an implicit method is a numerical approach that involves solving a system of algebraic equations at each time step. Unlike explicit methods, implicit methods do not update the solution directly from known values at the current step. Instead, they require solving a set of equations derived from the discretization of the governing equations.

The implicit nature of these methods allows larger time steps and increased stability compared to explicit ones, making them useful for steady-state or slowly-evolving problems. A common implicit method is the Crank-Nicolson scheme, which combines information from the current and next steps. This method offers improved numerical stability but is computationally more demanding. The choice between explicit and implicit methods depends on the specific problem characteristics and simulation requirements.

4. Discretization of Differential Equations

For the Prandtl boundary layer equations, the algebraic equations resulting from discretization of the differential equations (1.1) and (1.2) are analyzed below.

4.1 Explicit Method

For the term $\frac{\partial u}{\partial x}$, there are three discretization options: forward, backward, and central. However, in this specific case, we start from an initial velocity profile $u(x_i, y)$ and move along the flow direction. At $x = x_i$, the approximation for $\frac{\partial u}{\partial x}$using the backward method would require information from the point $x = x_i$-Δx, which does not exist. For this reason, the forward difference is used

$$\left(\frac{\partial u}{\partial x}\right)_{j,i} \approx \frac{u[j, i+1] - u[j, i]}{\Delta x}$$

For the derivative $\frac{\partial u}{\partial y}$, the central difference around the point (i, j) will be used again, and it will be given by

$$\left(\frac{\partial u}{\partial y}\right)_{j,i} \approx \frac{u[j+1, i] - u[j-1, i]}{2\Delta y}$$

Finally, for the second-order derivative $\frac{\partial^2 u}{\partial y^2}$, the corresponding finite difference equation is

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_{j,i} \approx \frac{u[j+1, i] - 2u[j, i] + u[j-1, i]}{\Delta y^2}$$

By substituting the above equations into equation (1.1), the final algebraic equation for the velocity equation is obtained as follows

$$u[j, i]\frac{u[j, i+1] - u[j, i]}{\Delta x} + v[j, i]\frac{u[j+1, i] - u[j-1, i]}{2\Delta y} = v\frac{u[j+1, i] - 2u[j, i] + u[j-1, i]}{\Delta y^2}$$

With appropriate rearrangements of the above equation, an expression for the velocity component $u_{i+1,j}$ at the next point is obtained

$$u[j,i+1] = u[j,i] + v\frac{u[j+1,i] - 2u[j,i] + u[j-1,i]}{\Delta y^2}\frac{\Delta x}{\Delta y^2} - \frac{u[j+1,i] - u[j-1,i]}{2}\frac{\Delta x}{\Delta y}\frac{v[j,i]}{u[j,i]} \quad (4.1.1)$$

Once the quantity $u[j,i+1]$ has been computed, the velocity vat the point $(j,i+1)$ can be determined from the continuity equation (1.2). An initial estimation would be to use again a central difference to approximate the velocity v, that is

$$\left(\frac{\partial v}{\partial y}\right)_{j,i} \approx \frac{v[j+1,i] - v[j-1,i]}{2\Delta y} \quad (4.1.2)$$

However, the term $v[j,i+1]$ does not appear so that it can be used in the next step. Therefore, we can use a backward difference, since the velocity v is known one step back in i, and not one step forward. Hence

$$\left(\frac{\partial v}{\partial y}\right)_{j,i+1} \approx \frac{v[j,i+1] - v[j-1,i+1]}{\Delta y}, \text{with } v[0,i+1] = 0 \quad (4.1.3)$$

This equation can be interpreted as a central difference evaluated at a midpoint between two grid points $(j,i+1)$ and $(j-1,i+1)$. This approach is used to achieve better derivative accuracy. Therefore

$$\left(\frac{\partial u}{\partial x}\right)_{j,i+1} \approx \frac{1}{2}\left[\frac{u[j,i+1] - u[j,i]}{\Delta x} + \frac{u[j-1,i+1] - u[j-1,i]}{\Delta x}\right] \quad (4.1.4)$$

By substituting the two equations above into equation (1.2), we obtain

$$\frac{1}{2}\left[\frac{u[j,i+1] - u[j,i]}{\Delta x} + \frac{u[j-1,i+1] - u[j-1,i]}{\Delta x}\right] + \frac{v[j,i+1] - v[j-1,i+1]}{\Delta y} = 0$$

and finally

$$v[j,i+1] = v[j-1,i+1] - \frac{1}{2}\left[\frac{u[j,i+1] - u[j,i]}{\Delta x} + \frac{u[j-1,i+1] - u[j-1,i]}{\Delta x}\right]\Delta y \quad (4.1.5)$$

Finally, it should be emphasized that the discretization step for each axis should not be arbitrary. A stability criterion applied for this specific case is

$$\Delta x \leq \frac{\frac{1}{2} u_{j,i} (\Delta y)^2}{v} \tag{4.1.6}$$

4.2 Implicit Method

Using this method, the goal is to create a system of algebraic equations to determine the boundary layer equations above the plate. As in the explicit method, the term $(\frac{\partial u}{\partial x})$ in equation (1.1) is computed using the forward difference method as follows:

$$\left(\frac{\partial u}{\partial x}\right)_{j,i} \approx \frac{u_{j,i+1} - u_{j,i}}{\Delta x} \tag{4.2.1}$$

However, in this case, for the other differential terms, the Crank-Nicolson method is used. Specifically

$$\left(\frac{\partial^2 u}{\partial y^2}\right)_{j,i} \approx \frac{1}{2}\left[\frac{u_{j+1,i+1} - 2u_{j,i+1} + u_{j-1,i+1}}{\Delta y^2} + \frac{u_{j+1,i} - 2u_{j,i} + u_{j-1,i}}{\Delta y^2}\right] \tag{4.2.2}$$

and

$$\left(\frac{\partial u}{\partial y}\right)_{j,i} \approx \frac{1}{2}\left[\frac{u_{j+1,i+1} - u_{j-1,i+1}}{2\Delta y} + \frac{u_{j+1,i} - u_{j-1,i}}{2\Delta y}\right] \tag{4.2.3}$$

Therefore, equation (1.1) becomes:

$$\begin{aligned} u[j,i]\frac{u[j,i+1] - u[j,i]}{\Delta x} &+ \frac{1}{2} v[j,i]\left[\frac{u[j+1,i+1] - u[j-1,i+1]}{2\Delta y} + \frac{u[j+1,i] - u[j-1,i]}{2\Delta y}\right] \\ &= \frac{1}{2} v\left[\frac{u[j+1,i+1] - 2u[j,i+1] + u[j-1,i+1]}{\Delta y^2}\right. \\ &+ \left.\frac{u[j+1,i] - 2u[j,i] + u[j-1,i]}{\Delta y^2}\right] \end{aligned}$$

By defining the coefficients

$$a = u_{j,i}\Delta x, b = \frac{v\Delta y^2}{2}, c = v_{j,i}\Delta y$$

This leads to

$$a(u_{j,i+1} - u_{j,i}) + \frac{1}{4}c(u_{j+1,i+1} - u_{j-1,i+1} + u_{j+1,i} - u_{j-1,i})$$
$$= \frac{1}{2}b(u_{j+1,i+1} - 2u_{j,i+1} + u_{j-1,i+1} + u_{j+1,i} - 2u_{j,i} + u_{j-1,i})$$

and finally

$$-\left(\frac{b}{2}+\frac{c}{4}\right)u_{j-1,i+1} + (b+a)u_{j,i+1} + \left(-\frac{b}{2}+\frac{c}{4}\right)u_{j+1,i+1} = \left(\frac{b}{2}+\frac{c}{4}\right)u_{j-1,i} + (-b+a)u_{j,i} + \left(\frac{b}{2}-\frac{c}{4}\right)u_{j+1,i} \quad (4.2.4)$$

where

$$k_i = (\frac{b}{2}+\frac{c}{4})u_{j-1,i} + (-b+a)u_{j,i} + (\frac{b}{2}-\frac{c}{4})u_{j+1,i}$$

Assume the spatial domain is temporarily discretized into 7 points. Since the two endpoints along the y-axis satisfy the boundary conditions (velocity at $y = 0$ is zero and at the top is the free-stream velocity), and k_2' and k_6' are known, analyzing the above equation results in the matrix form:

$$\begin{bmatrix} b+a & -\frac{b}{2}+\frac{c}{4} & 0 & 0 & 0 \\ -\frac{b}{2}-\frac{c}{4} & b+a & -\frac{b}{2}+\frac{c}{4} & 0 & 0 \\ 0 & -\frac{b}{2}-\frac{c}{4} & b+a & -\frac{b}{2}+\frac{c}{4} & 0 \\ 0 & 0 & -\frac{b}{2}-\frac{c}{4} & b+a & -\frac{b}{2}+\frac{c}{4} \\ 0 & 0 & 0 & -\frac{b}{2}-\frac{c}{4} & b+a \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{bmatrix} = \begin{bmatrix} k_2' \\ k_3 \\ k_4 \\ k_5 \\ k_6' \end{bmatrix} \quad (4.2.5)$$

The coefficient matrix is tridiagonal, and using the Thomas algorithm, the system can be efficiently solved.

5. Algorithm Development

5.1 Explicit Method

Initially, two (N_y, N_x) matrices of grid points filled with zeros are defined, representing the velocity profiles u and v. The initial conditions are set, meaning that the flow velocity uat the edge $x = 0$ and along the y-axis is equal to 2, while the boundary conditions are defined such that the velocity uon the plate $(y = 0)$ is zero, and the velocity above the boundary layer is 2.

Then, two nested for loops are created to calculate the velocity profiles, starting from the point $(2,2)$ and ending at the point corresponding to the free-stream region. For each value along the x-axis, the velocities along y are computed, i.e., $u(2,2)$, $u(2,3)$, $u(2,4)$, and so on. Once one iteration is completed, the process continues to the next point along the x-axis. The iterative procedure concludes and using equations (4.1.1) and (4.1.5), the velocity profiles are now fully determined.

5.2 Implicit Method

Initially, two matrices of size (Ny, Nx) are defined with zeros, representing the velocity profiles u and v. The initial conditions are set, specifically the flow velocity uat the inlet $x = 0$ along the y-axis, which is set equal to 2. The boundary conditions are also defined: the velocity uon the plate $(y = 0)$ is zero, while the velocity above the boundary layer is 2.

Two nested for loops are then formed, and the velocity profiles are computed starting from the initial point (2,2) up to the free-stream region. For each value along the x-axis, a tridiagonal matrix is computed, corresponding to the internal grid points along the y-axis, along with the vector of variables k from equation (4.2.5). Once the coefficients a, b, and c, which vary with each iteration along y, are computed, the matrix A and the variables k are calculated. In this way, for each grid point along x, the velocity profiles uand vare obtained via algebraic equations.

6. Grid Independence Study

Below is the grid independence study for both methods. The boundary layer thickness δ at the mid-point of the plate was chosen for various numbers of grid points. Convergence is observed after approximately 600 points along the y-axis.

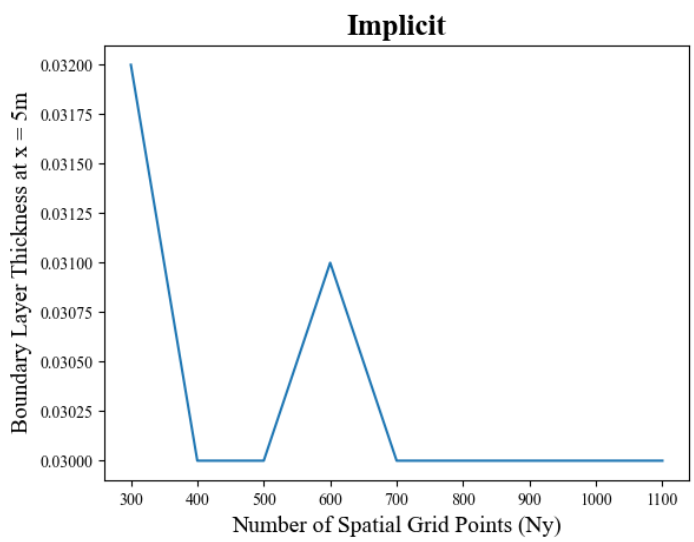

Figure 6.1. *Convergence using Implicit method*

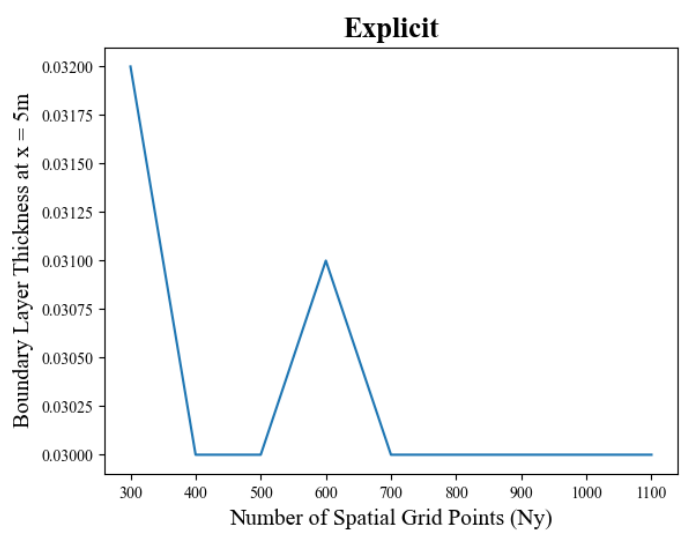

Figure 6.2. *Convergence using Explicit method*

7. Results Comparison

The quantities under study, which result from the analytical solution, were discussed in Section 2. Regarding the values obtained from the numerical solution, both in the explicit and implicit methods, the same formulas were used:

- For the boundary layer thickness δ, the point where the velocity reaches 99% of the free-stream velocity is identified from the data.
- For the displacement thickness δ_1, the following holds

$$\delta_1 = \int_0^{y_1} \left(1 - \frac{u}{U_\infty}\right) dy \tag{7.1}$$

- For the momentum thickness δ_2, the following holds

$$\delta_2 = \int_0^{y_1} \frac{u}{U_\infty}\left(1 - \frac{u}{U_\infty}\right) dy \tag{7.2}$$

Thus, for $\delta_1$and δ_2, the desired quantities can be obtained by replacing the integral with a summation using the sum command.

Regarding the friction coefficient for the numerical solution, the wall shear stress near the wall is required. This can be computed from its definition as

$$\tau = \mu \frac{\partial u}{\partial y} \approx \frac{u_2 - u_1}{\Delta y} \tag{7.3}$$

However, for higher accuracy, the following formula can be used

$$\tau_w = \mu \frac{\partial u}{\partial y} \approx \frac{-3u_1 + 4u_2 - u_3}{2\Delta y} \tag{7.4}$$

Finally, the skin friction coefficient $C_f$is calculated as

$$C_f = \frac{\tau_w}{0.5U_\infty^2} \tag{7.5}$$

Next, comparative plots between the Blasius solution and the explicit and implicit numerical solutions are presented.

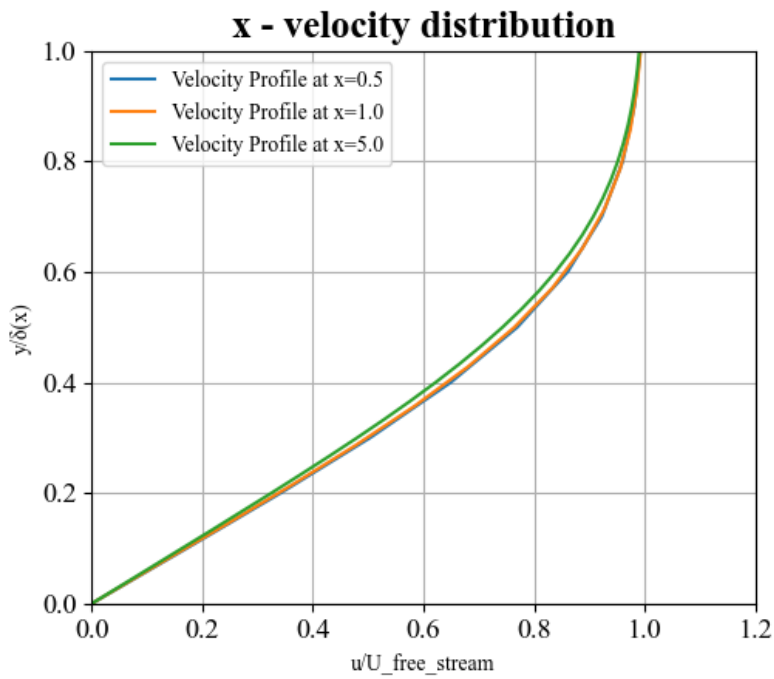

Figure 7.1. *Velocity distribution with Explicit method*

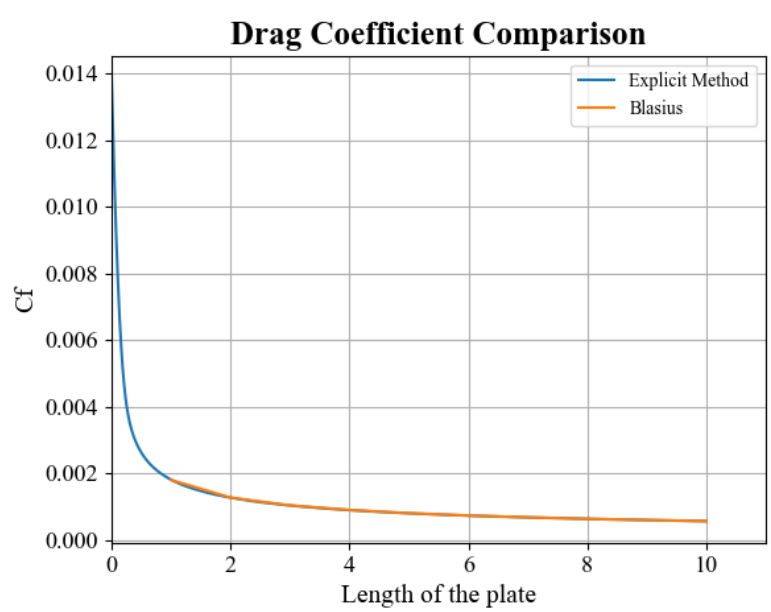

Figure 7.2. *Drag coefficient comparison between Explicit Method and the Blasius Solution*

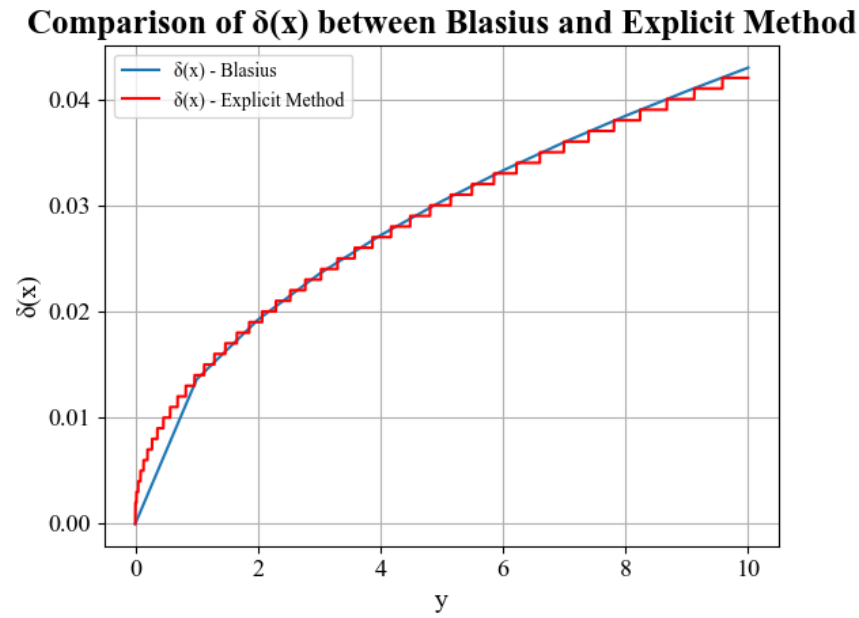

Figure 7.3. *Boundary layer thickness comparison between Explicit Method and the Blasius Solution*

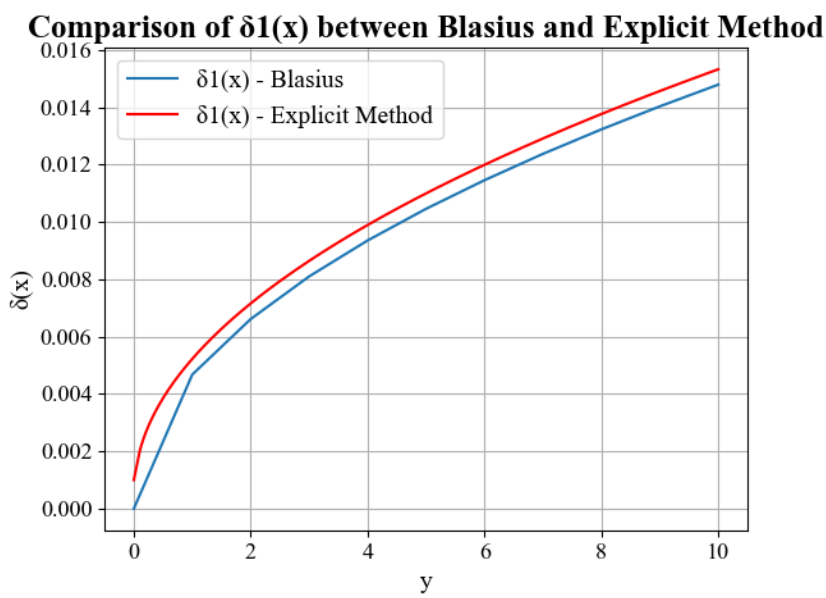

Figure 7.4. *Displacement thickness comparison between Explicit Method and the Blasius Solution*

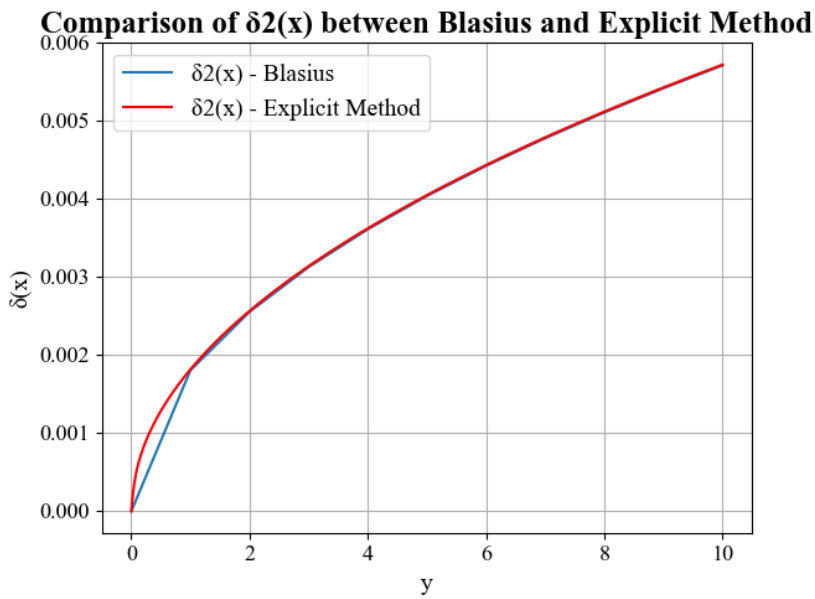

Figure 7.5. *Momentum thickness comparison between Explicit Method and the Blasius Solution*

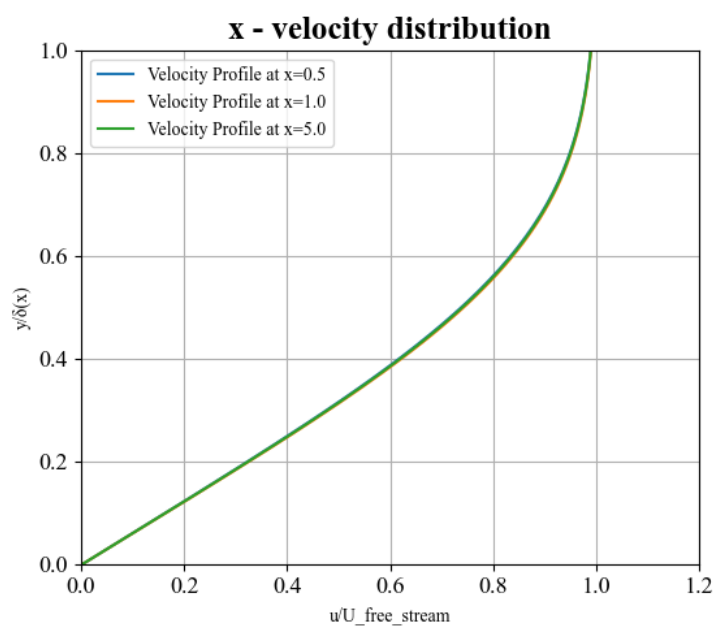

Figure 7.6. *Velocity distribution with Implicit method*

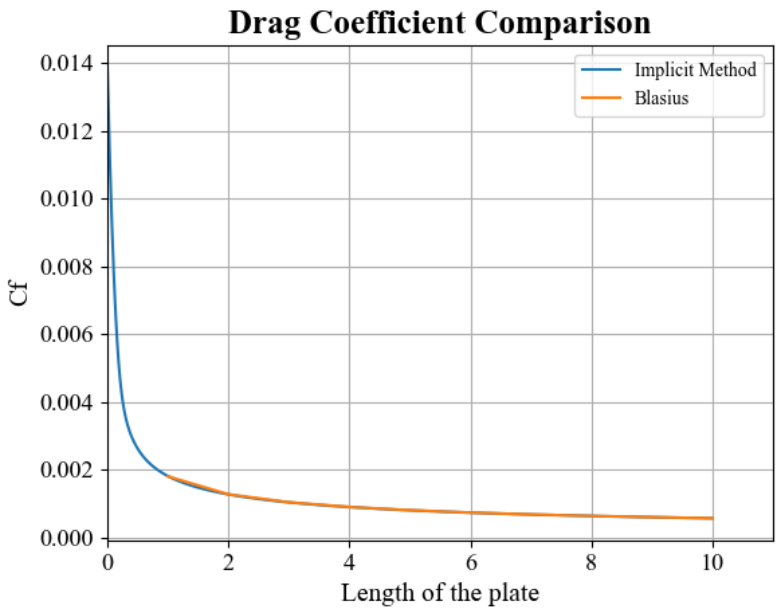

Figure 7.7. *Drag coefficient comparison between Implicit Method and the Blasius Solution*

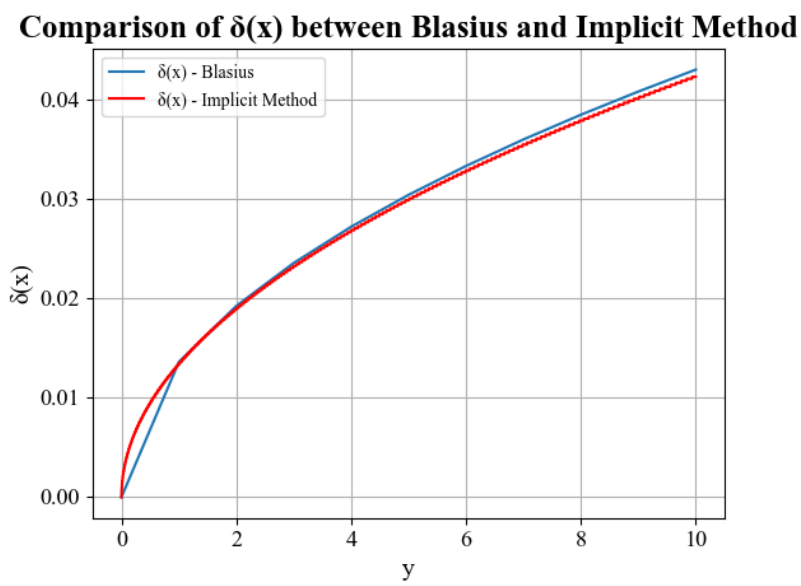

Figure 7.8. *Boundary layer thickness comparison between Implicit Method and the Blasius Solution*

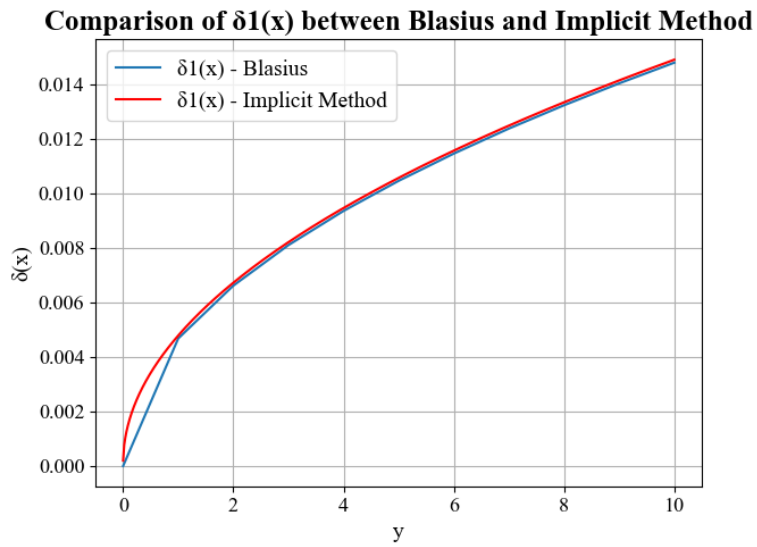

Figure 7.9. *Displacement thickness comparison between Implicit Method and the Blasius Solution*

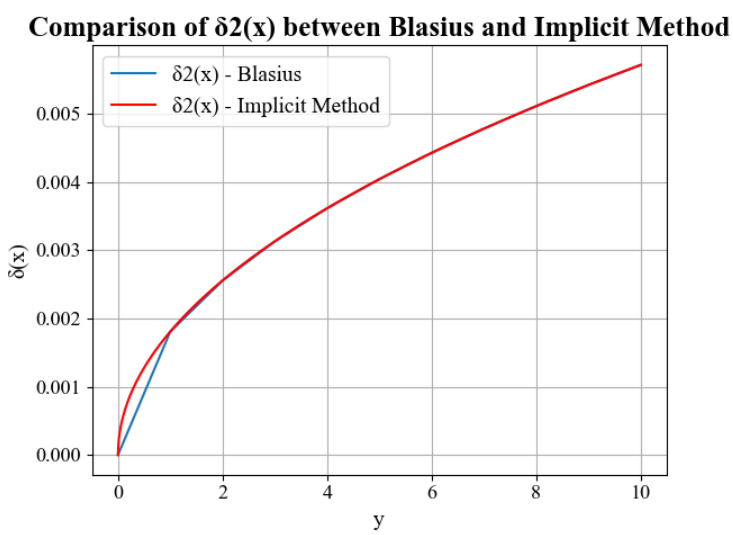

Figure 7.10. *Momentum thickness comparison between Implicit Method and the Blasius Solution*

From the above, it is clear that there is a high accuracy between the analytical values and both numerical methods. In the case of the Explicit method, it is more sensitive to the step size, and there is a possibility that the solution may diverge or even fail to converge. However, it is a more straightforward method and does not require as much computational power. In the case of the Implicit method, the solution always converges to a value, regardless of the step size. The only influence of the step size in this method is on the absolute accuracy of the solution. Nevertheless, due to the system of algebraic equations in the Implicit method, higher computational resources are required.

8. References

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